Excerpt 1

The purpose of deductive logic is to find relationships between statements (called premises and conclusions) that guarantee the truth of the conclusions if the premises are true. On the one hand, this means that we must be very careful to scrutinize the premises carefully. False premises can lead to false conclusions even when the logic of the argument is valid. (Logicians use the word "argument" to mean a set of premises leading to a conclusion.) On the other hand, the ability to spot bad logic ensures that we don't get fooled into accepting false conclusions based on obviously true premises. In fact, if we can spot bad logic, then we don't even need to look very carefully at the premises; we already know the conclusions are suspect at best. A valid logical argument, based on well-supported premises, leads to a trustworthy conclusion. A few years ago, when I was exhorting one of my classes to look for logical flaws (along with undocumented assertions, errors of fact, internal contradictions, and so on), one of my students said "It's all a matter of opinion." But not everything is simply a matter of opinion. There are clear and well-defined rules worked out by logicians that can be used to analyze arguments. Recent work in logic is all symbolic, essentially reducing arguments to formulas and examining the conditions under which the formulas are valid. Although this work is interesting, I'll focus here on the older verbal tradition in logic, which goes back to Aristotle.

A central element of this tradition is the syllogism. A syllogism is a form of logical argument that consists of two premises and a conclusion. The premises might be general propositions, taken to be always universally true; particular propositions about a single object, person, event, and so on; or conditional propositions concerning the circumstances under which a statement is true. In a valid syllogism, the truth of the two premises ensures without fail that the conclusion is true. Logicians customarily illustrate the use of the syllogism with simple examples like this:

(1) All cats are cute. {major premise}

(2) Smokey is a cat. {minor premise}

(3) Therefore, Smokey is cute. {conclusion}

It's quite obvious here that statements (1) and (2) cannot be true and yet have statement (3) false. People may have differing opinions about statement (1), and statement (2) is a simple matter of fact, either right or wrong. But if we accept both of these statements, the truth of statement (3) inevitably follows. Syllogisms like this one, which contain no conditional propositions, are called categorical syllogisms. The syllogism is a useful tool in constructing an argument. Once you have established a valid syllogism, you can then concentrate on establishing the validity of your premises. Perhaps more importantly, in analyzing someone else's argument, you can disentangle the logic from the premises. After you isolate the syllogistic form of the logic, you may see clearly that the reasoning is invalid. In that case, you don't have to worry about analyzing the premises (often a difficult and problematic task, leading to no unambiguous result). The veracity of the premises doesn't matter if the logic is invalid; the argument is no good anyway.

Another useful point concerning syllogisms is that the logic can be reversed. Once a valid syllogism has been established, one of the rules of logic tells us the following: If the conclusion is known to be false, then at least one of the premises must be false. Apply this reasoning to our example. If Smokey is not cute, then either Smokey isn't a cat or else not all cats are cute. I should also mention that there are some conclusions which cannot be drawn from syllogisms. For example, the truth of the conclusion doesn't tell us anything about the truth or falseness of the premises. Similarly, the falseness of the premises tells us nothing about the truth or falseness of the conclusion. Finally, let's look briefly at another important part of deductive logic, the conditional proposition. One type of conditional proposition tells us something about the truth of statements that are paired together (example: either 1+1 = 2 or my understanding of arithmetic is wrong). A second important type is the hypothetical proposition, which takes the form of an "if…then" assertion (example: if the Orioles win one more game, then they will be in the playoffs). These conditional propositions can also be used as premises in a syllogism. Gregory N Derry. What science is and how it works. Princeton University Press, 1999.

Excerpt 2

In deductive logic, our conclusions are based on a set of premises, and the truth of the premises implies the truth of the conclusions. The advantage of deductive logic is that we have the certainty of truth in those cases where the method can be used. The disadvantage of deductive logic is that we seldom have any well-defined general premises that we know are true. Instead, we can use a different form of reasoning, called inductive logic. The method of inductive logic is to use the truth of many particular statements to make a generalization, which is our conclusion. If every cat I've ever seen is cute, then I conclude based on this experience that all cats are cute. Obviously, the disadvantage of reasoning by induction (as opposed to deduction) is that my conclusions are less certain. I might run into an ugly cat tomorrow.

The use of inductive reasoning at some point is almost unavoidable. The difficulty is how to assess the validity of an inductive argument. Since a proof by induction can never be absolutely certain, how can we judge the quality of any conclusions drawn? Philosophers have put a lot of effort into deciding how to make such judgments; but for analyzing typical arguments, we can get a lot of use just from common sense. If a generalization is based on only one or two examples, then the conclusion is basically worthless. If the generalization is based on thousands of well-controlled and highly documented cases, then we can (at least provisionally) accept a conclusion in this case. In between these extremes, we'll accord conclusions the respect they deserve based on the amount of inductive evidence presented.

Gregory N Derry. What science is and how it works. Princeton University Press, 1999.